

Relational Schema

Missing memberApplicationID, memberStatus,

EMPLOYEE (employeeID, password)

PK: employeeID

ADMIN (employeeID, staffName, adminEmail, adminSex, adminDOB, employeeID)

PK: employeeID

MEMBERS(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary, memberRegistrationID)

PK: employeeID

LOAN(loanApplicationID, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAccount)

PK: loanApplicationID

LOANAPPROVAL(staffID, loanApplicationID, loanApproveDate, loanStatus)

PK: staffID, loanApplicationID

FK: staffID references ADMIN(staffID)

FK: loanApplicationID references LOAN(loanApplicationID)

GUARANTORDETAILS (icGuarantor, loanApplicationID, guarantorName, memberIdGuarantor, telGuarantor)

PK: icGuarantor, loanApplicationID

FK: loanApplicationID references LOAN (loanApplicationID)

APPLICATIONDETAILS(loanApplicationID, employeeID, loanApplicationDate, loanStatus)

PK: loanApplicationID, employeeID

FK: loanApplicationID references LOAN (loanApplicationID)

FK: employeeID references MEMBER (employeeID)

FINANCIALSTATUS (accountID, statementID, shareCapital, feeCapital, fixedSaving, memberFund, memberSaving, alBai, allnnah, bPulihKenderaan, roadRaxInsurance, specialScheme, specialSeasonCarnival, alQuadrulHassan, dateUpdated)

PK: accountID

TRANSACTION(transID, employeeID, accountID, transType, transAmt, transDate)

PK: transID, employeeID, accountID

FK: employeeID references MEMBERS(employeeID)

FK: accountID references FINANCIALSTATUS(accountID)

FEESSANDCONTRIBUTION (employeeID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others)

PK: employeeID

FK: employeeID references MEMBERS(employeeID)

FAMILYMEMBERINFO (icFamilyMember, employeeID, relationship, familyMemberName)
PK: icFamilyMember, employeeID
FK: employeeID references MEMBERS(employeeID)

ADDRESS(employeeID, address, postcode, state)
PK: employeeID
FK: employeeID references MEMBERS(employeeID)

Relational Schema (01/23)

- Missing memberRegistration, report, reportDetails, officeAddress, homeAddress, memberApplicationDetails, guarantor (DONE)
- Remove address (DONE)

(Updated according to 0115 Logical ERD)

EMPLOYEE (employeeID, password)
PK: employeeID

ADMIN (employeeID, staffName)
PK: employeeID

MEMBER(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, statementID)
PK: employeeID

MEMBERREGISTRATION (memberRegistrationID)
PK: memberRegistrationID

MEMBERAPPLICATIONDETAILS (employeeID, memberRegistrationID, regisDate, regisStatus)
PK: employeeID, memberRegistrationID
FK: employeeID references MEMBER(employeeID)
FK: memberRegistrationID references MEMBERREGISTRATION(memberRegistrationID)

MEMBERAPPROVAL(employeeID, memberRegistrationID, regisApproveDate, regisStatus)
PK: employeeID, memberRegistrationID
FK: employeeID references ADMIN(employeeID)
FK: memberRegistrationID references MEMBERREGISTRATION(memberRegistrationID)

LOAN(loanApplicationID, amountRequested, financingPeriod, monthlyInstallments, bankNo, bankName, financialType)
PK: loanApplicationID

LOANAPPROVAL(employeeID, loanApplicationID, loanApproveDate, loanStatus)

PK: employeeID, loanApplicationID

FK: employeeID references ADMIN(employeeID)

FK: loanApplicationID references LOAN(loanApplicationID)

APPLICATIONDETAILS (loanApplicationID, employeeID, applicationDate, loanStatus)

PK: loanApplicationID, employeeID

FK: loanApplicationID references LOAN(loanApplicationID)

FK: employeeID references MEMBER(employeeID)

GUARANTOR (icGuarantor, loanApplicationID, guarantorName, memberIDGuarantor, guarantorPF, telGuarantor)

PK: icGuarantor, loanApplicationID

FK: loanApplicationID references LOAN (loanApplicationID)

FINANCIALSTATUS (accountID, statementID, memberSaving, alBai, allnnah, bPulihKenderaan, roadTaxInsurance, specialScheme, alQadrulHassan, dateUpdated)

PK: accountID

TRANSACTION(transID, employeeID, accountID, transType, transAmt, transDate)

PK: transID, employeeID, accountID

FK: employeeID references MEMBER(employeeID)

FK: accountID references FINANCIALSTATUS(accountID)

FEESANDCONTRIBUTION (employeeID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others)

PK: employeeID

FK: employeeID references MEMBER(employeeID)

FAMILYMEMBERINFO (icFamilyMember, employeeID, relationship, familyMemberName)

PK: icFamilyMember, employeeID

FK: employeeID references MEMBER(employeeID)

HOMEADDRESS (employeeID, homeAddress, homePostcode, homeState)

PK: employeeID

FK: employeeID references MEMBER(employeeID)

OFFICEADDRESS (employeeID, officeAddress, officePostcode, officeState)

PK: employeeID

FK: employeeID references MEMBER(employeeID)

REPORT(employeeID, loanApplicationID, accountID, reportType, reportFormat)

PK: employeeID, loanApplicationID, accountID

FK: employeeID references MEMBER(employeeID)

FK: loanApplicationID references LOAN(loanApplicationID)

FK: accountID references FINANCIALSTATUS(accountID)

REPORTDETAILS(employeeID, generateDate, reportStatus)

PK: employeeID

FK: employeeID references ADMIN(employeeID)

Normalization

1. employee - admin

1NF:

Employee_Admin (employeeID, password, staffName, adminEmail, adminSex, adminDOB)

PK: employeeID

2NF:

Admin (employeeID, staffName, adminEmail, adminSex, adminDOB)

PK: employeeID

Employee (employeeID, password)

PK: employeeID

FK: employeeID references Admin(employeeID)

3NF:

Admin (employeeID, staffName, adminEmail, adminSex, adminDOB)

PK: employeeID

Employee (employeeID, password)

PK: employeeID

FK: employeeID references Admin(employeeID)

2. familyMemberInfo - member

1NF:

familyMemberInfo (relationship, familyMemberName, icFamilyMember, employeeID)

PK: icFamilyMember

FK: employeeID references Employee_Member (employeeID)

2NF:

familyMemberInfo (relationship, familyMemberName, icFamilyMember, employeeID)

PK: icFamilyMember

FK: employeeID references Employee_Member (employeeID)

3NF:

familyMemberInfo (relationship, familyMemberName, icFamilyMember, employeeID)

PK: icFamilyMember

FK: employeeID references Employee_Member (employeeID)

3. employee - member

1NF:

Employee_Member(employeeID, password, employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, address, position, officeAddress, monthlySalary, memberRegistrationID)

PK: employeeID, employeeID

2NF:

Person(ic, name, email, phoneNumber, sex, religion, maritalStatus, nation, address)

PK: ic

Employee(employeeID, password, position, officeAddress, monthlySalary)

PK: employeeID

Member(employeeID, memberName, noPF, memberRegistrationID)

PK: employeeID

Employee_Member(employeeID, employeeID)

PK: employeeID, employeeID

FK: employeeID references Employee(employeeID)

FK: employeeID references Member(employeeID)

3NF: Doesn't exist

4. admin - member

1NF:

Admin_Member(employeeID, staffName, adminEmail, adminSex, adminDOB, employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, address, position, officeAddress, monthlySalary, memberRegistrationID)

PK: employeeID, employeeID

2NF:

Admin(employeeID, staffName, adminEmail, adminSex, adminDOB)

PK: employeeID

Member(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, address, position, officeAddress, monthlySalary, memberRegistrationID)

PK: employeeID

Admin_Member(employeeID, employeeID)

PK: employeeID, employeeID

FK: employeeID references Admin(employeeID)

FK: employeeID references Member(employeeID)

3NF: Doesn't exist

5. admin - loanApproval

1NF:

admin_loanApproval(employeeID, staffName, adminEmail, adminSex, adminDOB, loanApproveDate, loanStatus)

PK: employeeID

2NF:

admin(employeeID, staffName, adminEmail, adminSex, adminDOB)

PK: employeeID

admin_loanApproval(employeeID, loanApproveDate, loanStatus)

PK: employeeID

FK: employeeID references admin (employeeID)

3NF: Doesn't exist

6. member - feesAndContribution

1NF:

members_feesAndContribution(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, address, position, officeAddress, monthlySalary, memberRegistrationID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others)

PK: employeeID

2NF:

member(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, address, position, officeAddress, monthlySalary, memberRegistrationID)

PK: employeeID

member_feesAndContribution(employeeID, email, phoneNumber, sex, religion, maritalStatus, nation, address, position, officeAddress, monthlySalary, memberRegistrationID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others)

PK: employeeID

FK: employeeID references member(employeeID)

3NF:

memberRegister(memberRegistrationID, memberName, ic)

PK: memberRegistrationID

member_feesAndContribution(employeeID, email, phoneNumber, sex, religion, maritalStatus, nation, address, position, officeAddress, monthlySalary, memberRegistrationID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others)

PK: employeeID

FK: employeeID references member(employeeID)

FK: memberRegistrationID references memberRegister(memberRegistrationID)

7. member - financialStatus

1NF:

member_financialStatus(employeeID, accountID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, address, position, officeAddress, monthlySalary, memberRegistrationID, statementID, memberSaving, alBai, alInnah, bPulihKenderaan, roadTaxInsurance, specialScheme, alQadrulHassan, dateUpdated)

PK: employeeID, accountID

2NF:

member(employeeID, noPF, memberName, ic)

PK: employeeID

member_financialStatus(employeeID, accountID, email, phoneNumber, sex, religion, maritalStatus, nation, address, position, officeAddress, monthlySalary, memberRegistrationID, statementID, memberSaving, alBai, alInnah, bPulihKenderaan, roadTaxInsurance, specialScheme, alQadrulHassan, dateUpdated)

PK: employeeID, accountID

FK: employee references member(employeeID)

3NF: Doesn't exist

8. loan - guarantorDetails

1NF:

loan_guarantorDetails(loanApplicationID, icGuarantor, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAccount, guarantorName, memberIdGuarantor, telGuarantor)

PK: loanApplicationID, icGuarantor

2NF:

guarantor(icGuarantor, guarantorName, memberIdGuarantor, telGuarantor)

PK: icGuarantor

loan_guarantorDetails(loanApplicationID, icGuarantor, amountRequested, financingPeriod, monthlyInstallments, bankName, bankAccount)

PK: loanApplicationID, icGuarantor

FK: icGuarantor references guarantor (icGuarantor)

3NF:

Does not exist.

9. Member - applicationDetails

1NF:

member_applicationDetails(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, add1, postcode1, state1, position, add2, postcode2, state2, monthlySalary, memberRegistrationID, applicationDate, memberStatus)

2NF:

member_applicationDetails(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, add1, postcode1, state1, position, add2, postcode2, state2, monthlySalary, memberRegistrationID, applicationDate, memberStatus)

3NF:

member_applicationDetails(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, add1, postcode1, state1, position, add2, postcode2, state2, monthlySalary, memberRegistrationID, applicationDate, memberStatus)

homeAddress (employeeID, add1, postcode1, state1)

PK: employeeID

FK: employeeID references member(employeeID)

officeAddress (employeeID, add2, postcode2, state2)

PK: employeeID

FK: employeeID references member(employeeID)

10. Member - transaction

1NF:

member_transaction (employeeID, transID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, add1, postcode1, state1, position, add2, postcode2, state2, monthlySalary, memberRegistrationID, transType, transAmt, transDate)

PK: employeeID

PK: transID

2NF:

member(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, add1, postcode1, state1, position, add2, postcode2, state2, monthlySalary, memberRegistrationID)

PK: employeeID

transaction(employeeID, transID, transType, transAmt, transDate)

PK: employeeID, transID

FK: employeeID references member(employeeID)

3NF:

member(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, monthlySalary, memberRegistrationID)

PK: employeeID

transaction(employeeID, transID, transType, transAmt, transDate)

PK: employeeID, transID

FK: employeeID references member(employeeID)

address(employeeID, addNo, add, postcode, state)

PK: employeeID, addNo

FK: employeeID references member(employeeID)

11. Loan- applicationDetails

1NF:

Loan_Application (loanApplicationID, amountRequested, financingPeriod, monthlyInstallments, bankNo, bankName, bankHolderName , loanApplicationDate, loanStatus)

PK: loanApplicationID

2NF:

Loan (loanApplicationID, amountRequested, financingPeriod, monthlyInstallments)

PK: loanApplicationID

Bank (bankNo, bankName, bankHolderName)

PK: bankNo

Loan_Application (loanApplicationID, bankNo, loanApplicationDate, loanStatus)

PK: loanApplicationID, bankNo

FK: loanApplicationID reference Loan(loanApplicationID)

FK: bankNo reference Bank(bankNo)

3NF:

Loan (loanApplicationID, amountRequested, financingPeriod, monthlyInstallment)

PK: loanApplicationID

Bank (bankNo, bankName, bankHolderName)

PK: bankNo

Loan_Application (loanApplicationID, bankNo, loanApplicationDate, loanStatus)

PK: loanApplicationID, bankNo

FK: loanApplicationID reference Loan(loanApplicationID)

FK: bankNo reference Bank(bankNo)

12. Loan- loanApproval

1NF:

Loan_Approval (loanApplicationID, amountRequested, financingPeriod, monthlyInstallments, bankNo, bankName, bankHolderName , loanApproveDate, loanStatus)
PK: loanApplicationID

2NF:

Loan (loanApplicationID, amountRequested, financingPeriod, monthlyInstallment) PK:
loanApplicationID

Bank (bankNo, bankName, bankHolderName)
PK: bankNo

Loan_Approval (loanApplicationID, bankNo, loanApproveDate, loanStatus)
PK: loanApplicationID
FK: loanApplicationID reference Loan (loanApplicationID)
FK: bankNo reference Bank(bankNo)

3NF:

Loan (loanApplicationID, amountRequested, financingPeriod, monthlyInstallment)
PK: loanApplicationID

Bank (bankNo, bankName, bankHolderName)
PK: bankNo

Loan_Approval (loanApplicationID, bankNo, loanApproveDate, loanStatus)
PK: loanApplicationID, bankNo
FK: loanApplicationID reference Loan(loanApplicationID)
FK: bankNo reference Bank(bankNo)

13. Member - address

1NF:

Member_address(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary, address, postcode, state)

PK:employeeID

2NF:

Member(employeeID, noPF, memberName, ic, email, phoneNumber, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

Address(employeeID, address, postcode, state)

PK:employeeID

FK:employeeID references member(employeeID)

14. memberRegistration-feesAndContribution

1NF:

memberRegistration(memberRegistrationID)

PK: memberRegistrationID

feeAnd Contribution(entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others, memberRegistrationID)

PK: memberRegistrationID

FK: memberRegistrationID references memberRegistration (memberRegistrationID)

2NF:

memberRegistration(memberRegistrationID)

PK: memberRegistrationID

feeAnd Contribution(entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others, memberRegistrationID)

PK: memberRegistrationID

FK: memberRegistrationID references memberRegistration (memberRegistrationID)

3NF:

memberRegistration(memberRegistrationID)

PK: memberRegistrationID

feeAnd Contribution(entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others, memberRegistrationID)

PK: memberRegistrationID

FK: memberRegistrationID references memberRegistration (memberRegistrationID)

15. Admin - memberApproval

1NF:

admin_memberApproval(employeeID, staffName, regisApproveDate, regisStatus)

2NF:

admin(employeeID, staffName)

memberApproval(regisApproveDate, regisStatus)

16. familyMemberInfo - member

1NF:

familyMemberInfo (relationship, familyMemberName, icFamilyMember, employeeID)

PK: icFamilyMember

FK: employeeID references Employee_Member (employeeID)

2NF:

familyMemberInfo (relationship, familyMemberName, icFamilyMember, employeeID)

PK: icFamilyMember

FK: employeeID references Employee_Member (employeeID)

3NF:

familyMemberInfo (relationship, familyMemberName, icFamilyMember, employeeID)

PK: icFamilyMember

FK: employeeID references Employee_Member (employeeID)

Normalization

1. employee - admin

1NF:

Employee_Admin (employeeID, password, staffName)

PK: employeeID

2NF:

Employee (employeeID, password)

PK: employeeID

Admin (employeeID, staffName)

PK: employeeID

FK: employeeID references Employee(employeeID)

3NF:

Employee (employeeID, password)

PK: employeeID

Admin (employeeID, staffName)

PK: employeeID

FK: employeeID references Employee(employeeID)

2. admin - loanApproval

1NF:

admin_loanApproval(employeeID, staffName, loanApproveDate, loanStatus)

PK: employeeID

2NF:

admin_loanApproval(employeeID, staffName, loanApproveDate, loanStatus)

PK: employeeID

3NF:

admin_loanApproval(employeeID, staffName, loanApproveDate, loanStatus)

PK: employeeID

3. Loan - loanApproval

1NF:

loanApproval(loanApplicationID, amountRequested, financingPeriod, monthlyInstallments, bankNo, bankName, loanApproveDate, loanStatus)

PK: loanApplicationID

2NF:

loanApproval(loanApplicationID, amountRequested, financingPeriod, monthlyInstallments, bankNo, bankName, loanApproveDate, loanStatus)

PK: loanApplicationID

3NF:

loan(loanApplicationID, amountRequested, financingPeriod, monthlyInstallment)

PK: loanApplicationID

bank(bankNo, bankName)

PK: bankNo

loanApproval (loanApplicationID, bankNo, loanApproveDate, loanStatus)

PK: loanApplicationID, bankNo

FK: loanApplicationID reference loan(loanApplicationID)

FK: bankNo reference bank(bankNo)

4. loan - guarantorDetails

1NF:

loan_guarantorDetails(loanApplicationID, icGuarantor, amountRequested, financingPeriod, monthlyInstallments, bankNo, bankName, guarantorName, memberIdGuarantor, telGuarantor)

PK: loanApplicationID, icGuarantor

2NF:

loan(loanApplicationID, amountRequested, financingPeriod, monthlyInstallment)

PK: loanApplicationID

guarantor(icGuarantor, guarantorName, memberIdGuarantor, telGuarantor)

PK: icGuarantor

loan_guarantorDetails(loanApplicationID, icGuarantor, bankNo, bankName)

PK: loanApplicationID, icGuarantor

FK: icGuarantor references guarantor (icGuarantor)

FK: loanApplicationID references loan (loanApplicationID)

3NF:

loan(loanApplicationID, amountRequested, financingPeriod, monthlyInstallment)

PK: loanApplicationID

guarantor(icGuarantor, guarantorName, memberIdGuarantor, telGuarantor)

PK: icGuarantor

bank(bankNo, bankName)

PK: bankNo

loanGuarantorDetails(loanApplicationID, icGuarantor)

PK: loanApplicationID, icGuarantor

FK: icGuarantor references guarantor (icGuarantor)

FK: loanApplicationID references loan (loanApplicationID)

5. Loan- applicationDetails

1NF:

Loan_Application (loanApplicationID, amountRequested, financingPeriod,
monthlyInstallments, bankNo, bankName, loanApplicationDate, loanStatus)
PK: loanApplicationID

2NF:

Loan_Application (loanApplicationID, amountRequested, financingPeriod,
monthlyInstallments, bankNo, bankName, loanApplicationDate, loanStatus)
PK: loanApplicationID

3NF:

Loan (loanApplicationID, amountRequested, financingPeriod, monthlyInstallments)
PK: loanApplicationID

Bank (bankNo, bankName)
PK: bankNo

LoanApplication (loanApplicationID, bankNo, loanApplicationDate, loanStatus)
PK: loanApplicationID, bankNo
FK: loanApplicationID reference Loan(loanApplicationID)
FK: bankNo reference Bank(bankNo)

6. admin - member

1NF:

Admin_Member(employeeID, staffName, noPF, memberName, ic, email, phoneNumber,
phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)
PK: employeeID

2NF:

Admin(employeeID, staffName)
PK: employeeID

Member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex,
religion, maritalStatus, nation, position, monthlySalary)
PK: employeeID

Admin_Member(employeeID)
PK: employeeID
FK: employeeID references Admin(employeeID)
FK: employeeID references Member(employeeID)

3NF:

Admin(employeeID, staffName)
PK: employeeID

Member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex,
religion, maritalStatus, nation, position, monthlySalary)
PK: employeeID

Admin_Member(employeeID)
PK: employeeID
FK: employeeID references Admin(employeeID)
FK: employeeID references Member(employeeID)

7. employee - member

1NF:

Employee_Member(employeeID, password, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

2NF:

Employee(employeeID, password)

PK: employeeID

Member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

Employee_Member(employeeID)

PK: employeeID

FK: employeeID references Employee(employeeID)

FK: employeeID references Member(employeeID)

3NF:

Employee(employeeID, password)

PK: employeeID

Member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

Employee_Member(employeeID)

PK: employeeID

FK: employeeID references Employee(employeeID)

FK: employeeID references Member(employeeID)

8. Member - applicationDetails

1NF:

memberApplicationDetails(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary, applicationDate, loanStatus)

2NF:

memberApplicationDetails(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary, applicationDate, loanStatus)

3NF:

memberApplicationDetails(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary, applicationDate, loanStatus)

9. Member - transaction

1NF:

member_transaction (employeeID, transID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary, transType, transAmt, transDate)

PK: employeeID

PK: transID

2NF:

member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

transaction(transID, transType, transAmt, transDate)

PK: transID

Member_transaction (employeeID, transID)

PK: employeeID, transID

FK: employeeID references member(employeeID)

FK: transID references transaction(transID)

3NF:

member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

transaction(transID, transType, transAmt, transDate)

PK: transID

Member_transaction (employeeID, transID)

PK: employeeID, transID

FK: employeeID references member(employeeID)

FK: transID references transaction(transID)

10. member - financialStatus

1NF:

member_financialStatus(employeeID, accountID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary, statementID, memberSaving, alBai, alInnah, bPulihKenderaan, roadTaxInsurance, specialScheme, alQadrulHassan, dateUpdated)

PK: employeeID, accountID

2NF:

member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

financialStatus(accountID, statementID, memberSaving, alBai, alInnah, bPulihKenderaan, roadTaxInsurance, specialScheme, alQadrulHassan, dateUpdated)

PK: accountID

member_financialStatus(employeeID, accountID)

PK: employeeID, accountID

FK: employee references member(employeeID)

FK: accountID references financialStatus(accountID)

3NF:

member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

financialStatus(accountID, statementID, memberSaving, alBai, alInnah, bPulihKenderaan, roadTaxInsurance, specialScheme, alQadrulHassan, dateUpdated)

PK: accountID

member_financialStatus(employeeID, accountID)

PK: employeeID, accountID

FK: employee references member(employeeID)

FK: accountID references financialStatus(accountID)

11. Transaction - financialStatus

1NF:

transaction_financialStatus (transID, accountID, transType, transAmt, transDate, statementID, memberSaving, alBai, alInnah, bPulihKenderaan, roadTaxInsurance, specialScheme, alQadrulHassan, dateUpdated)

PK: transID, accountID

2NF:

transaction (transID, transType, transAmt, transDate)

PK: transID

financialStatus(accountID, statementID, memberSaving, alBai, alInnah, bPulihKenderaan, roadTaxInsurance, specialScheme, alQadrulHassan, dateUpdated)

PK: accountID

transaction_financialStatus (transID, accountID)

PK: transID, accountID

FK: transID references transaction(transID)

FK: accountID references financialStatus(accountID)

3NF:

transaction (transID, transType, transAmt, transDate)

PK: transID

financialStatus(accountID, statementID, memberSaving, alBai, alInnah, bPulihKenderaan, roadTaxInsurance, specialScheme, alQadrulHassan, dateUpdated)

PK: accountID

transaction_financialStatus (transID, accountID)

PK: transID, accountID

FK: transID references transaction(transID)

FK: accountID references financialStatus(accountID)

12. Member - homeAddress

1NF:

Member_homeAddress(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary, homeAddress, homePostcode, homeState)

PK: employeeID

2NF:

Member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

member_homeAddress(employeeID, homeAddress, homePostcode, homeState)

PK: employeeID

FK: employeeID references member(employeeID)

3NF:

Member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

member_homeAddress(employeeID, homeAddress, homePostcode, homeState)

PK: employeeID

FK: employeeID references member(employeeID)

13. Member - officeAddress

1NF:

Member_officeAddress(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary, officeAddress, officePostcode, officeState)

PK: employeeID

2NF:

Member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

member_officeAddress(employeeID, officeAddress, officePostcode, officeState)

PK: employeeID

FK: employeeID references member(employeeID)

3NF:

Member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

member_officeAddress(employeeID, officeAddress, officePostcode, officeState)

PK: employeeID

FK: employeeID references member(employeeID)

14. Admin - memberApproval

1NF:

admin_memberApproval(employeeID, staffName, regisApproveDate, regStatus)

PK: employeeID

2NF:

admin_memberApproval(employeeID, staffName, regisApproveDate, regStatus)

PK: employeeID

3NF:

admin_memberApproval(employeeID, staffName, regisApproveDate, regStatus)

PK: employeeID

15. memberApproval - memberRegistration

1NF:

memberApproval_memberRegistration(memberRegistrationID, regisApproveDate, regisStatus)

PK: memberRegistrationID

2NF:

memberApproval_memberRegistration(memberRegistrationID, regisApproveDate, regisStatus)

PK: memberRegistrationID

3NF:

memberApproval_memberRegistration(memberRegistrationID, regisApproveDate, regisStatus)

PK: memberRegistrationID

16. memberRegistration - familyMemberInfo

1NF:

memberRegistration_familyMemberInfo(memberRegistrationID, icFamilyMember, relationship, familyMemberName)

PK: memberRegistrationID, icFamilyMember

2NF:

familyMemberInfo(icFamilyMember, relationship, familyMemberName)

PK: icFamilyMember

memberRegistration_familyMemberInfo(memberRegistrationID, icFamilyMember)
PK: memberRegistrationID, icFamilyMember
FK: icFamilyMember references familyMemberInfo(icFamilyMember)

3NF:

familyMemberInfo(icFamilyMember, relationship, familyMemberName) employeeID
PK: icFamilyMember
memberRegistration_familyMemberInfo(memberRegistrationID, icFamilyMember)
PK: memberRegistrationID, icFamilyMember
FK: icFamilyMember references familyMemberInfo(icFamilyMember)

17. memberRegistration - feesAndContribution

1NF:

memberRegistration_feesAndContribution(memberRegistrationID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others)
PK: memberRegistrationID

2NF:

memberRegistration_feesAndContribution(memberRegistrationID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others)
PK: memberRegistrationID

3NF:

memberRegistration_feesAndContribution(memberRegistrationID, entryFee, modalShare, feeCapital, deposit, contribution, fixedDeposit, others)
PK: memberRegistrationID

18. memberRegistration - memberApplicationDetails

1NF:

memberRegistration_memberApplicationDetails(memberRegistrationID, regisDate, regisStatus)
PK: memberRegistrationID

2NF:

memberRegistration_memberApplicationDetails(memberRegistrationID, regisDate, regisStatus)
PK: memberRegistrationID

3NF:

memberRegistration_memberApplicationDetails(memberRegistrationID, regisDate, regisStatus)
PK: memberRegistrationID

19. Member - memberApplicationDetails

1NF:

member_memberApplicationDetails(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary, regisDate, regisStatus)

PK: employeeID

2NF:

member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

member_memberApplicationDetails(employeeID, regisDate, regisStatus)

PK: employeeID

FK: employeeID references member(employeeID)

3NF:

member(employeeID, noPF, memberName, ic, email, phoneNumber, phoneHome, sex, religion, maritalStatus, nation, position, monthlySalary)

PK: employeeID

member_memberApplicationDetails(employeeID, regisDate, regisStatus)

PK: employeeID

FK: employeeID references member(employeeID)